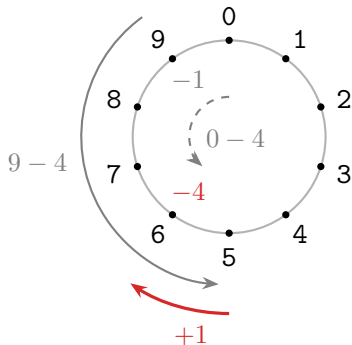


Fixed-width numbers live on a **circle**: counting past the largest value wraps back to 0. With one digit, 9 is one step before 0, so it acts as  $-1$ .



$$9 - 4 = 5$$

subtract from 9: never a borrow

$$5 + 1 = 6$$

add one

6 lands where  $0 - 4$  does, so 6 stands for  $-4$ .

$4 + 6 = 10$ , which wraps to 0

with two digits the circle has 100 steps:  $99 - 49 = 50$ , then  $+1$   
gives  $51 \equiv -49$